

NTA JEE Mains Jan 2026

Test Date	21/01/2026
Test Time	3:00 PM - 6:00 PM
Subject	B. Tech

Section : Mathematics Section A

Q.1

For the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -29 & 49 \\ -13 & 18 \end{bmatrix}$, if $(A^{15} + B) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, then among the following which one is true ?

Options

1. $x = 11, y = 2$
2. $x = 5, y = 7$
3. $x = 18, y = 11$
4. $x = 16, y = 3$

Question Type : MCQ

Question ID : 860654830

Option 1 ID : 8606542824

Option 2 ID : 8606542822

Option 3 ID : 8606542825

Option 4 ID : 8606542823

Q.2

Let $A = \{x : |x^2 - 10| \leq 6\}$ and $B = \{x : |x - 2| > 1\}$. Then

Options

1. $A - B = [2, 3)$
2. $B - A = (-\infty, -4) \cup (-2, 1) \cup (4, \infty)$
3. $A \cup B = (-\infty, 1] \cup (2, \infty)$
4. $A \cap B = [-4, -2] \cup [3, 4]$

Question Type : MCQ

Question ID : 860654833

Option 1 ID : 8606542834

Option 2 ID : 8606542837

Option 3 ID : 8606542836

Option 4 ID : 8606542835

Q.3 Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16, 16)$. If $P(\alpha, \beta)$ divides this focal chord internally in the ratio $5 : 2$, then the minimum value of $\alpha + \beta$ is equal to :

Options

1. 22
2. 16
3. 7
4. 5

Question Type : MCQ

Question ID : 860654836

Option 1 ID : 8606542849

Option 2 ID : 8606542848

Option 3 ID : 8606542847

Option 4 ID : 8606542846

Q.4 Let $f(x) = x^3 + x^2 f'(1) + 2x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then the value of $f'(5)$ is :

Options

1. $\frac{62}{5}$
2. $\frac{657}{5}$
3. $\frac{117}{5}$
4. $\frac{2}{5}$

Question Type : MCQ

Question ID : 860654842

Option 1 ID : 8606542871

Option 2 ID : 8606542872

Option 3 ID : 8606542873

Option 4 ID : 8606542870

Q.5

Let $y = y(x)$ be the solution of the differential equation $\sec x \frac{dy}{dx} - 2y = 2 + 3\sin x$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$y(0) = -\frac{7}{4}$. Then $y\left(\frac{\pi}{6}\right)$ is equal to :

Options

1. $-\frac{5}{2}$
2. $-\frac{5}{4}$
3. $-3\sqrt{3} - 7$
4. $-3\sqrt{2} - 7$

Question Type : **MCQ**

Question ID : **860654844**

Option 1 ID : **8606542881**

Option 2 ID : **8606542880**

Option 3 ID : **8606542878**

Option 4 ID : **8606542879**

Q.6

Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate

axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

Options

1. **16**
2. **6**
3. **10**
4. **12**

Question Type : **MCQ**

Question ID : **860654839**

Option 1 ID : **8606542861**

Option 2 ID : **8606542858**

Option 3 ID : **8606542859**

Option 4 ID : **8606542860**

Q.7 If the system of equations

$$3x + y + 4z = 3$$

$$2x + \alpha y - z = -3$$

$$x + 2y + z = 4$$

has no solution, then the value of α is equal to :

Options

1. 4
2. 19
3. 23
4. 13

Question Type : **MCQ**

Question ID : **860654831**

Option 1 ID : **8606542826**

Option 2 ID : **8606542828**

Option 3 ID : **8606542829**

Option 4 ID : **8606542827**

Q.8 Let α and β be the roots of the equation $x^2 + 2ax + (3a + 10) = 0$ such that $\alpha < 1 < \beta$. Then the set of all possible values of a is :

Options

1. $(-\infty, -2) \cup (5, \infty)$
2. $(-\infty, \frac{-11}{5}) \cup (5, \infty)$
3. $(-\infty, \frac{-11}{5})$
4. $(-\infty, -3)$

Question Type : **MCQ**

Question ID : **860654829**

Option 1 ID : **8606542818**

Option 2 ID : **8606542821**

Option 3 ID : **8606542820**

Option 4 ID : **8606542819**

Q.9 The positive integer n , for which the solutions of the equation

$$x(x+2) + (x+2)(x+4) + \cdots + (x+2n-2)(x+2n) = \frac{8n}{3}$$
 are two consecutive even integers, is :

Options

1. 12
2. 3
3. 6
4. 9

Question Type : **MCQ**

Question ID : **860654828**

Option 1 ID : **8606542817**

Option 2 ID : **8606542814**

Option 3 ID : **8606542815**

Option 4 ID : **8606542816**

Q.10

A random variable X takes values 0, 1, 2, 3 with probabilities $\frac{2a+1}{30}$, $\frac{8a-1}{30}$, $\frac{4a+1}{30}$, b respectively,

where $a, b \in \mathbb{R}$. Let μ and σ respectively be the mean and standard deviation of X such that $\sigma^2 + \mu^2 = 2$.

Then $\frac{a}{b}$ is equal to :

Options

1. 60
2. 12
3. 30
4. 3

Question Type : **MCQ**

Question ID : **860654835**

Option 1 ID : **8606542842**

Option 2 ID : **8606542844**

Option 3 ID : **8606542843**

Option 4 ID : **8606542845**

Q.11 Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbf{R}$ and $f'(a-1) = 0$, where a is a real number. Let $g(x) = f(\tan^2 x - 2 \tan x + a)$, $0 < x < \frac{\pi}{2}$.

Consider the following two statements :

(I) g is increasing in $\left(0, \frac{\pi}{4}\right)$

(II) g is decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

Then,

Options

1. Neither (I) nor (II) is True
2. Only (I) is True
3. Both (I) and (II) are True
4. Only (II) is True

Question Type : **MCQ**

Question ID : **860654843**

Option 1 ID : **8606542877**

Option 2 ID : **8606542874**

Option 3 ID : **8606542876**

Option 4 ID : **8606542875**

Q.12 Let $y^2 = 12x$ be the parabola with its vertex at O. Let P be a point on the parabola and A be a point on the x-axis such that $\angle OPA = 90^\circ$. Then the locus of the centroid of such triangles OPA is :

Options

1. $y^2 - 9x + 6 = 0$
2. $y^2 - 2x + 8 = 0$
3. $y^2 - 6x + 4 = 0$
4. $y^2 - 4x + 8 = 0$

Question Type : **MCQ**

Question ID : **860654837**

Option 1 ID : **8606542852**

Option 2 ID : **8606542853**

Option 3 ID : **8606542851**

Option 4 ID : **8606542850**

Q.13 Let z be the complex number satisfying $|z - 5| \leq 3$ and having maximum positive principal argument.

Then $34 \left| \frac{5z - 12}{5iz + 16} \right|^2$ is equal to :

Options

1. 16
2. 20
3. 12
4. 26

Question Type : MCQ

Question ID : 860654827

Option 1 ID : 8606542811

Option 2 ID : 8606542812

Option 3 ID : 8606542810

Option 4 ID : 8606542813

Q.14 Let $A = \{2, 3, 5, 7, 9\}$. Let R be the relation on A defined by xRy if and only if $2x \leq 3y$. Let l be the number of elements in R , and m be the minimum number of elements required to be added in R to make it a symmetric relation. Then $l + m$ is equal to :

Options

1. 21
2. 27
3. 23
4. 25

Question Type : MCQ

Question ID : 860654826

Option 1 ID : 8606542806

Option 2 ID : 8606542809

Option 3 ID : 8606542807

Option 4 ID : 8606542808

Q.15 For a triangle ABC , let $\vec{p} = \vec{BC}$, $\vec{q} = \vec{CA}$ and $\vec{r} = \vec{BA}$. If $|\vec{p}| = 2\sqrt{3}$, $|\vec{q}| = 2$ and $\cos\theta = \frac{1}{\sqrt{3}}$,

where θ is the angle between $\vec{p}$ and $\vec{q}$, then $|\vec{p} \times (\vec{q} - 3\vec{r})|^2 + 3|\vec{r}|^2$ is equal to :

Options

1. 200
2. 410
3. 220
4. 340

Question Type : MCQ

Question ID : 860654841

Option 1 ID : 8606542866

Option 2 ID : 8606542869

Option 3 ID : 8606542867

Option 4 ID : 8606542868

Q.16 If the area of the region $\{(x, y) : 1 - 2x \leq y \leq 4 - x^2, x \geq 0, y \geq 0\}$ is $\frac{\alpha}{\beta}$, $\alpha, \beta \in \mathbf{N}$, $\gcd(\alpha, \beta) = 1$, then the value of $(\alpha + \beta)$ is :

Options

1. 91
2. 67
3. 85
4. 73

Question Type : MCQ

Question ID : 860654845

Option 1 ID : 8606542885

Option 2 ID : 8606542882

Option 3 ID : 8606542884

Option 4 ID : 8606542883

Q.17 If the line $\alpha x + 4y = \sqrt{7}$, where $\alpha \in \mathbf{R}$, touches the ellipse $3x^2 + 4y^2 = 1$ at the point P in the first quadrant, then one of the focal distances of P is :

Options

1. $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{11}}$
2. $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{7}}$
3. $\frac{1}{\sqrt{3}} + \frac{1}{2\sqrt{5}}$
4. $\frac{1}{\sqrt{3}} - \frac{1}{2\sqrt{5}}$

Question Type : MCQ

Question ID : 860654838

Option 1 ID : 8606542857

Option 2 ID : 8606542856

Option 3 ID : 8606542854

Option 4 ID : 8606542855

Q.18

Let the line L_1 be parallel to the vector $-3\hat{i} + 2\hat{j} + 4\hat{k}$ and pass through the point $(2, 6, 7)$, and the line L_2 be parallel to the vector $2\hat{i} + \hat{j} + 3\hat{k}$ and pass through the point $(4, 3, 5)$. If the line L_3 is parallel to the vector $-3\hat{i} + 5\hat{j} + 16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $|\vec{CD}|^2$ is equal to :

Options

1. 89
2. 171
3. 312
4. 290

Question Type : **MCQ**

Question ID : **860654840**

Option 1 ID : **8606542862**

Option 2 ID : **8606542863**

Option 3 ID : **8606542865**

Option 4 ID : **8606542864**

Q.19

The largest $n \in \mathbb{N}$, for which 7^n divides $101!$, is :

Options

1. 19
2. 18
3. 15
4. 16

Question Type : **MCQ**

Question ID : **860654834**

Option 1 ID : **8606542838**

Option 2 ID : **8606542839**

Option 3 ID : **8606542841**

Option 4 ID : **8606542840**

Q.20 Let $a_1, \frac{a_2}{2}, \frac{a_3}{2^2}, \dots, \frac{a_{10}}{2^9}$ be a G.P. of common ratio $\frac{1}{\sqrt{2}}$. If $a_1 + a_2 + \dots + a_{10} = 62$, then a_1 is equal to :

Options

1. $\sqrt{2} - 1$
2. $2 - \sqrt{2}$
3. $2(2 - \sqrt{2})$
4. $2(\sqrt{2} - 1)$

Question Type : MCQ

Question ID : 860654832

Option 1 ID : 8606542831

Option 2 ID : 8606542833

Option 3 ID : 8606542832

Option 4 ID : 8606542830

Section : Mathematics Section B

Q.21 If $\left(\frac{1}{{}^{15}C_0} + \frac{1}{{}^{15}C_1}\right)\left(\frac{1}{{}^{15}C_1} + \frac{1}{{}^{15}C_2}\right) \dots \left(\frac{1}{{}^{15}C_{12}} + \frac{1}{{}^{15}C_{13}}\right) = \frac{\alpha^{13}}{{}^{14}C_0 {}^{14}C_1 \dots {}^{14}C_{12}}$, then 30α is equal to _____.

Question Type : SA

Question ID : 860654846

Q.22 If P is a point on the circle $x^2 + y^2 = 4$, Q is a point on the straight line $5x + y + 2 = 0$ and $x - y + 1 = 0$ is the perpendicular bisector of PQ, then 13 times the sum of abscissa of all such points P is _____.

Question Type : SA

Question ID : 860654847

Q.23 If $\int_0^1 4 \cot^{-1}(1 - 2x + 4x^2) dx = a \tan^{-1}(2) - b \log_e(5)$, where $a, b \in \mathbb{N}$, then $(2a + b)$ is equal to _____.

Question Type : SA

Question ID : 860654850

Q.24 Let the maximum value of $(\sin^{-1}x)^2 + (\cos^{-1}x)^2$ for $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to _____.

Question Type : SA

Question ID : 860654848

Q.25 Let $[\cdot]$ denote the greatest integer function and $f(x) = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n \left\lfloor \frac{k^2}{3^x} \right\rfloor$. Then $12 \sum_{j=1}^{\infty} f(j)$ is equal to _____.

Question Type : SA

Question ID : 860654849

Q.26 The energy of an electron in an orbit of the Bohr's atom is $-0.04E_0$ eV where E_0 is the ground state energy. If L is the angular momentum of the electron in this orbit and h is the Planck's constant, then

$\frac{2\pi L}{h}$ is _____ :

Options

1. 6
2. 5
3. 4
4. 2

Question Type : MCQ

Question ID : 860654869

Option 1 ID : 8606542966

Option 2 ID : 8606542963

Option 3 ID : 8606542964

Option 4 ID : 8606542965

Q.27 The kinetic energy of a simple harmonic oscillator is oscillating with angular frequency of

176 rad/s. The frequency of this simple harmonic oscillator is _____ Hz. $\left[\text{take } \pi = \frac{22}{7} \right]$

Options

1. 28
2. 14
3. 88
4. 176

Question Type : MCQ

Question ID : 860654860

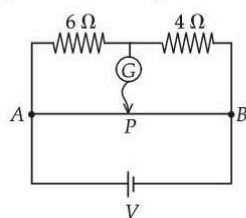
Option 1 ID : 8606542928

Option 2 ID : 8606542929

Option 3 ID : 8606542930

Option 4 ID : 8606542927

- Q.28** The total length of potentiometer wire AB is 50 cm in the arrangement as shown in figure. If P is the point where the galvanometer shows zero reading then the length AP is _____ cm.



Options

1. 30
2. 15
3. 25
4. 20

Question Type : MCQ

Question ID : 860654863

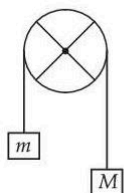
Option 1 ID : 8606542939

Option 2 ID : 8606542941

Option 3 ID : 8606542942

Option 4 ID : 8606542940

- Q.29** The pulley shown in figure is made using a thin rim and two rods of length equal to diameter of the rim. The rim and each rod have a mass of M . Two blocks of mass of M and m are attached to two ends of a light string passing over the pulley, which is hinged to rotate freely in vertical plane about its center. The magnitudes of the acceleration experienced by the blocks is _____ (assume no slipping of string on pulley).



Options

1. $\frac{(M-m)g}{M+m}$
2. $\frac{(M-m)g}{2M+m}$
3. $\frac{(M-m)g}{\left[\left(\frac{8}{3}\right)M+m\right]}$
4. $\frac{(M-m)g}{\left[\left(\frac{13}{6}\right)M+m\right]}$

Question Type : MCQ

Question ID : 860654854

Option 1 ID : 8606542905

Option 2 ID : 8606542906

Option 3 ID : 8606542904

Option 4 ID : 8606542903

Q.30 A battery with EMF E and internal resistance r is connected across a resistance R . The power consumption in R will be maximum when :

Options

1. $R = 2r$
2. $R = r$
3. $R = \sqrt{2} r$
4. $R = r/2$

Question Type : **MCQ**

Question ID : **860654866**

Option 1 ID : **8606542953**

Option 2 ID : **8606542952**

Option 3 ID : **8606542954**

Option 4 ID : **8606542951**

Q.31 Surface tension of two liquids (having same densities), T_1 and T_2 , are measured using capillary rise method utilizing two tubes with inner radii of r_1 and r_2 where $r_1 > r_2$. The measured liquid heights in these tubes are h_1 and h_2 respectively. [Ignore the weight of the liquid about the lowest point of meniscus]. The heights h_1 and h_2 and surface tensions T_1 and T_2 satisfy the relation :

Options

1. $h_1 > h_2$ and $T_1 < T_2$
2. $h_1 < h_2$ and $T_1 = T_2$
3. $h_1 > h_2$ and $T_1 = T_2$
4. $h_1 = h_2$ and $T_1 = T_2$

Question Type : **MCQ**

Question ID : **860654858**

Option 1 ID : **8606542920**

Option 2 ID : **8606542922**

Option 3 ID : **8606542921**

Option 4 ID : **8606542919**

Q.32 A spherical body of radius r and density σ falls freely through a viscous liquid having density ρ and viscosity η and attains a terminal velocity v_0 . Estimated maximum error in the quantity η is : (Ignore errors associated with σ , ρ and g , gravitational acceleration)

Options

1. $2 \left[\frac{\Delta r}{r} - \frac{\Delta v_0}{v_0} \right]$

2. $2 \frac{\Delta r}{r} - \frac{\Delta v_0}{v_0}$

3. $\frac{2\Delta r}{r} + \frac{\Delta v_0}{v_0}$

4. $2 \left[\frac{\Delta r}{r} + \frac{\Delta v_0}{v_0} \right]$

Question Type : **MCQ**

Question ID : **860654852**

Option 1 ID : **8606542896**

Option 2 ID : **8606542897**

Option 3 ID : **8606542898**

Option 4 ID : **8606542895**

Q.33 Keeping the significant figures in view, the sum of the physical quantities 52.01 m, 153.2 m and 0.123 m is :

Options

1. 205.3 m

2. 205.333 m

3. 205 m

4. 205.33 m

Question Type : **MCQ**

Question ID : **860654851**

Option 1 ID : **8606542892**

Option 2 ID : **8606542894**

Option 3 ID : **8606542891**

Option 4 ID : **8606542893**

Q.34 Given below are two statements :

Statement I : In a Young's double slit experiment, the angular separation of fringes will increase as the screen is moved away from the plane of the slits

Statement II : In a Young's double slit experiment, the angular separation of fringes will increase when monochromatic source is replaced by another monochromatic source of higher wavelength

In the light of the above statements, choose the **correct answer** from the options given below :

Options

1. **Statement I is true but Statement II is false**
2. **Statement I is false but Statement II is true**
3. **Both Statement I and Statement II are false**
4. **Both Statement I and Statement II are true**

Question Type : **MCQ**

Question ID : **860654867**

Option 1 ID : **8606542957**

Option 2 ID : **8606542958**

Option 3 ID : **8606542956**

Option 4 ID : **8606542955**

Q.35 A capacitor C is first charged fully with potential difference of V_0 and disconnected from the battery. The charged capacitor is connected across an inductor having inductance L . In t s 25% of the initial energy in the capacitor is transferred to the inductor. The value of t is _____s.

Options

1. $\pi\sqrt{\frac{LC}{2}}$
2. $\frac{\pi\sqrt{LC}}{3}$
3. $\frac{\pi\sqrt{LC}}{2}$
4. $\frac{\pi\sqrt{LC}}{6}$

Question Type : **MCQ**

Question ID : **860654861**

Option 1 ID : **8606542931**

Option 2 ID : **8606542933**

Option 3 ID : **8606542932**

Option 4 ID : **8606542934**

- Q.36** A body of mass 2 kg is moving along x -direction such that its displacement as function of time is given by $x(t) = at^2 + \beta t + \gamma$ m, where $a = 1 \text{ m/s}^2$, $\beta = 1 \text{ m/s}$ and $\gamma = 1 \text{ m}$. The work done on the body during the time interval $t = 2 \text{ s}$ to $t = 3 \text{ s}$, is _____ J.

Options

1. 49
2. 42
3. 24
4. 12

Question Type : MCQ

Question ID : 860654856

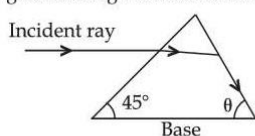
Option 1 ID : 8606542912

Option 2 ID : 8606542914

Option 3 ID : 8606542911

Option 4 ID : 8606542913

- Q.37** As shown in the diagram, when the incident ray is parallel to base of the prism, the emergent ray grazes along the second surface.



If refractive index of the material of prism is $\sqrt{2}$, the angle θ of prism is.

Options

1. 90°
2. 60°
3. 45°
4. 75°

Question Type : MCQ

Question ID : 860654868

Option 1 ID : 8606542962

Option 2 ID : 8606542960

Option 3 ID : 8606542959

Option 4 ID : 8606542961

- Q.38** Two cars A and B each of mass 10^3 kg are moving on parallel tracks separated by a distance of 10 m, in same direction with speeds 72 km/h and 36 km/h. The magnitude of angular momentum of car A with respect to car B is _____ J.s.

Options

1. 3×10^5
2. 10^5
3. 2×10^5
4. 3.6×10^5

Question Type : MCQ

Question ID : 860654855

Option 1 ID : 8606542908

Option 2 ID : 8606542907

Option 3 ID : 8606542910

Option 4 ID : 8606542909

Q.39 The r.m.s. speed of oxygen molecules at 47°C is equal to that of the hydrogen molecules kept at _____ $^{\circ}\text{C}$. (Mass of oxygen molecule/mass of hydrogen molecule = 32/2)

Options

1. - 253
2. - 20
3. - 235
4. - 100

Question Type : **MCQ**

Question ID : **860654859**

Option 1 ID : **8606542924**

Option 2 ID : **8606542923**

Option 3 ID : **8606542926**

Option 4 ID : **8606542925**

Q.40 A river of width 200 m is flowing from west to east with a speed of 18 km/h. A boat, moving with speed of 36 km/h in still water, is made to travel one-round trip (bank to bank of the river). Minimum time taken by the boat for this journey and also the displacement along the river bank are _____ and _____ respectively.

Options

1. 40 s and 200 m
2. 20 s and 100 m
3. 40 s and 100 m
4. 40 s and 0 m

Question Type : **MCQ**

Question ID : **860654853**

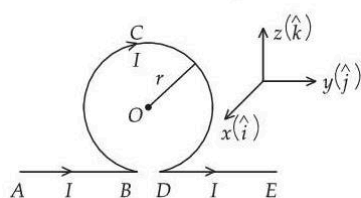
Option 1 ID : **8606542900**

Option 2 ID : **8606542899**

Option 3 ID : **8606542902**

Option 4 ID : **8606542901**

- Q.41** An infinitely long straight wire carrying current I is bent in a planer shape as shown in the diagram. The radius of the circular part is r . The magnetic field at the centre O of the circular loop is :



Options

1. $-\frac{\mu_0}{2\pi} \frac{I}{r} (\pi + 1) \hat{i}$
2. $-\frac{\mu_0}{2\pi} \frac{I}{r} (\pi - 1) \hat{i}$
3. $\frac{\mu_0}{2\pi} \frac{I}{r} (\pi + 1) \hat{i}$
4. $\frac{\mu_0}{2\pi} \frac{I}{r} (\pi - 1) \hat{i}$

Question Type : MCQ

Question ID : 860654864

Option 1 ID : 8606542943

Option 2 ID : 8606542944

Option 3 ID : 8606542946

Option 4 ID : 8606542945

- Q.42** Consider two identical metallic spheres of radius R each having charge Q and mass m . Their centers have an initial separation of $4R$. Both the spheres are given an initial speed of u towards each other. The minimum value of u , so that they can just touch each other is :

(Take $k = \frac{1}{4\pi\epsilon_0}$ and assume $kQ^2 > Gm^2$ where G is the Gravitational constant)

Options

1. $\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{2kQ^2} \right)}$
2. $\sqrt{\frac{kQ^2}{4mR} \left(1 - \frac{Gm^2}{kQ^2} \right)}$
3. $\sqrt{\frac{kQ^2}{4mR} \left(1 + \frac{Gm^2}{kQ^2} \right)}$
4. $\sqrt{\frac{kQ^2}{2mR} \left(1 - \frac{Gm^2}{kQ^2} \right)}$

Question Type : MCQ

Question ID : 860654865

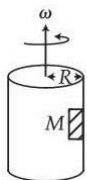
Option 1 ID : 8606542950

Option 2 ID : 8606542948

Option 3 ID : 8606542947

Option 4 ID : 8606542949

- Q.43** A large drum having radius R is spinning around its axis with angular velocity ω , as shown in figure. The minimum value of ω so that a body of mass M remains stuck to the inner wall of the drum, taking the coefficient of friction between the drum surface and mass M as μ , is :



Options

1. $\sqrt{\frac{\mu g}{R}}$
2. $\sqrt{\frac{2g}{\mu R}}$
3. $\sqrt{\frac{g}{\mu R}}$
4. $\sqrt{\frac{g}{2\mu R}}$

Question Type : **MCQ**

Question ID : **860654857**

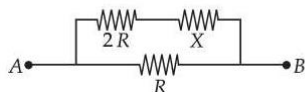
Option 1 ID : **8606542918**

Option 2 ID : **8606542917**

Option 3 ID : **8606542916**

Option 4 ID : **8606542915**

- Q.44** Two known resistances of $R \Omega$ and $2R \Omega$ and one unknown resistance $X \Omega$ are connected in a circuit as shown in the figure. If the equivalent resistance between points A and B in the circuit is $X \Omega$, then the value of X is _____ Ω .



Options

1. $(\sqrt{3} - 1)R$
2. R
3. $2(\sqrt{3} - 1)R$
4. $(\sqrt{3} + 1)R$

Question Type : **MCQ**

Question ID : **860654862**

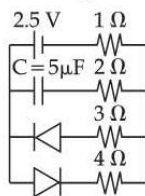
Option 1 ID : **8606542937**

Option 2 ID : **8606542935**

Option 3 ID : **8606542938**

Option 4 ID : **8606542936**

Q.45 The charge stored by the capacitor C in the given circuit in the steady state is _____ μC .



Options

1. 5
2. 10
3. 7.5
4. 12.5

Question Type : **MCQ**

Question ID : **860654870**

Option 1 ID : **8606542967**

Option 2 ID : **8606542968**

Option 3 ID : **8606542969**

Option 4 ID : **8606542970**

Section : **Physics Section B**

Q.46 A particle having electric charge $3 \times 10^{-19} \text{ C}$ and mass $6 \times 10^{-27} \text{ kg}$ is accelerated by applying an electric potential of 1.21 V. Wavelength of the matter wave associated with the particle is $\alpha \times 10^{-12} \text{ m}$. The value of α is _____. (Take Planck's constant $= 6.6 \times 10^{-34} \text{ J.s}$)

Question Type : **SA**

Question ID : **860654875**

Q.47 The terminal velocity of a metallic ball of radius 6 mm in a viscous fluid is 20 cm/s. The terminal velocity of another ball of same material and having radius 3 mm in the same fluid will be _____ cm/s.

Question Type : **SA**

Question ID : **860654871**

Q.48 A diatomic gas ($\gamma = 1.4$) does 100 J of work when it is expanded isobarically. Then the heat given to the gas _____ J.

Question Type : **SA**

Question ID : **860654872**

Q.49 An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10 \text{ mho/m}$. The ratio of maximum conduction current density to maximum displacement current density is _____.

$$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \right]$$

Question Type : **SA**

Question ID : **860654873**

- Q.50** In a Young's double slit experiment set up, the two slits are kept 0.4. mm apart and screen is placed at 1 m from slits. If a thin transparent sheet of thickness $20\text{ }\mu\text{m}$ is introduced in front of one of the slits then center bright fringe shifts by 20 mm on the screen. The refractive index of transparent sheet is given by $\frac{\alpha}{10}$, where α is _____.

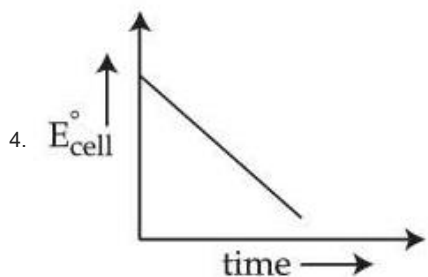
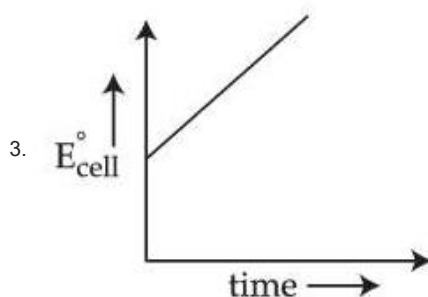
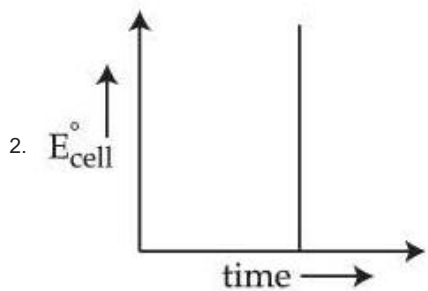
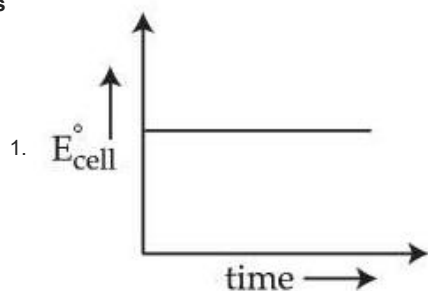
Question Type : **SA**

Question ID : **860654874**

Section : **Chemistry Section A**

- Q.51** For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature ?

Options



Question Type : **MCQ**

Question ID : **860654880**

Option 1 ID : **8606542995**

Option 2 ID : **8606542992**

Option 3 ID : **8606542993**

Option 4 ID : **8606542994**

Q.52 Match List - I with List - II.

List - I	List - II
Reagents	Reaction Name (Involving aldehydes)
A. H_2 , Pd-BaSO ₄	I. Etard Reaction
B. SnCl_2 , HCl	II. Rosenmund Reduction
C. CrO_2Cl_2 , CS ₂	III. Gatterman - Koch Reaction
D. CO, HCl, Anhyd. AlCl ₃	IV. Stephen Reaction

Choose the **correct** answer from the options given below :

Options

1. A-IV, B-I, C-II, D-III
2. A-IV, B-III, C-I, D-II
3. A-II, B-IV, C-I, D-III
4. A-II, B-III, C-IV, D-I

Question Type : **MCQ**

Question ID : **860654892**

Option 1 ID : **8606543040**

Option 2 ID : **8606543041**

Option 3 ID : **8606543042**

Option 4 ID : **8606543043**

Q.53 The correct increasing order of C – H(A), C – O(B), C = O(C) and C $\equiv$ N(D) bonds in terms of covalent bond length is :

Options

1. $\text{D} < \text{C} < \text{A} < \text{B}$
2. $\text{D} < \text{C} < \text{B} < \text{A}$
3. $\text{A} < \text{B} < \text{C} < \text{D}$
4. $\text{A} < \text{D} < \text{C} < \text{B}$

Question Type : **MCQ**

Question ID : **860654878**

Option 1 ID : **8606542987**

Option 2 ID : **8606542986**

Option 3 ID : **8606542984**

Option 4 ID : **8606542985**

- Q.54** By usual analysis, 1.00 g of compound (X) gave 1.79 g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is : (nearest integer)
(Given, molar mass in g mol^{-1} : O = 16, Mg = 24, P = 31)

Options

1. 40
2. 20
3. 50
4. 30

Question Type : MCQ

Question ID : 860654886

Option 1 ID : 8606543018

Option 2 ID : 8606543016

Option 3 ID : 8606543019

Option 4 ID : 8606543017

- Q.55** Given below are two statements :

Statement I : The correct order in terms of atomic/ionic radii is $\text{Al} > \text{Mg} > \text{Mg}^{2+} > \text{Al}^{3+}$.

Statement II : The correct order in terms of the magnitude of electron gain enthalpy is $\text{Cl} > \text{Br} > \text{S} > \text{O}$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. **Statement I** is false but **Statement II** is true
2. Both **Statement I** and **Statement II** are true
3. Both **Statement I** and **Statement II** are false
4. **Statement I** is true but **Statement II** is false

Question Type : MCQ

Question ID : 860654882

Option 1 ID : 8606543003

Option 2 ID : 8606543000

Option 3 ID : 8606543001

Option 4 ID : 8606543002

- Q.56** The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is :



Options

1. $\text{ClO}_4^- > \text{CH}_3\text{COO}^- > ^-\text{OH} > \text{PhO}^-$
2. $\text{CH}_3\text{COO}^- > \text{PhO}^- > ^-\text{OH} > \text{ClO}_4^-$
3. $\text{PhO}^- > ^-\text{OH} > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$
4. $^-\text{OH} > \text{PhO}^- > \text{CH}_3\text{COO}^- > \text{ClO}_4^-$

Question Type : MCQ

Question ID : 860654887

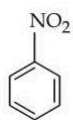
Option 1 ID : 8606543023

Option 2 ID : 8606543021

Option 3 ID : 8606543022

Option 4 ID : 8606543020

Q.57



- (1) $\text{Br}_2/\text{FeBr}_3/\Delta$
 (2) $\text{Sn}/\text{HCl}/\Delta$
 (3) pH neutralisation
 (4) $\text{Br}_2/\text{H}_2\text{O}$
 (5) NaNO_2/HBr , $0-5^\circ\text{C}$
 (6) CuBr/NaBr
- Major Product (P)

Consider the above sequence of reactions. The number of bromine atom(s) in the final product (P) will be :

Options

1. 1
2. 3
3. 5
4. 6

Question Type : MCQ

Question ID : 860654893

Option 1 ID : 8606543044

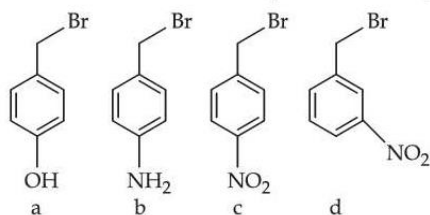
Option 2 ID : 8606543046

Option 3 ID : 8606543045

Option 4 ID : 8606543047

Q.58

The correct order of reactivity of the following benzyl halides towards reaction with KCN is :



Options

1. $b > a > d > c$
2. $a > b > c > d$
3. $a > b > d > c$
4. $b > a > c > d$

Question Type : MCQ

Question ID : 860654891

Option 1 ID : 8606543036

Option 2 ID : 8606543039

Option 3 ID : 8606543037

Option 4 ID : 8606543038

Q.59 Consider the following spectral lines for atomic hydrogen :

- A. First line of Paschen series
- B. Second line of Balmer series
- C. Third line of Paschen series
- D. Fourth line of Bracket series

The correct arrangement of the above lines in ascending order of energy is :

Options

- 1. $C < D < B < A$
- 2. $A < B < C < D$
- 3. $D < C < A < B$
- 4. $D < A < C < B$

Question Type : **MCQ**

Question ID : **860654877**

Option 1 ID : **8606542983**

Option 2 ID : **8606542980**

Option 3 ID : **8606542982**

Option 4 ID : **8606542981**

Q.60 Given below are some of the statements about Mn and Mn_2O_7 . Identify the correct statements.

- A. Mn forms the oxide Mn_2O_7 , in which Mn is in its highest oxidation state.
- B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn.
- C. Mn_2O_7 is an ionic oxide.
- D. The structure of Mn_2O_7 consists of one bridged oxygen.

Choose the **correct** answer from the options given below :

Options

- 1. A, C and D Only
- 2. A, B and D Only
- 3. A, B and C Only
- 4. A, B, C and D

Question Type : **MCQ**

Question ID : **860654884**

Option 1 ID : **8606543010**

Option 2 ID : **8606543009**

Option 3 ID : **8606543011**

Option 4 ID : **8606543008**

Q.61 On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are :

Options

1. CrO_2Cl_2 and +6
2. $Cr_2O_2Cl_2$ and +3
3. $Cr_2O_2Cl_2$ and +6
4. CrO_2Cl_2 and +5

Question Type : **MCQ**

Question ID : **860654895**

Option 1 ID : **8606543053**

Option 2 ID : **8606543054**

Option 3 ID : **8606543055**

Option 4 ID : **8606543052**

Q.62 Match List - I with List - II.

List - I

Pair of Compounds

- A. 2-Methylpropene and but-1-ene
- B. Cis-but-2-ene and trans-but-2-ene
- C. 2-Butanol and diethyl ether
- D. But-1-ene and but-2-ene

List - II

Type of Isomers

- I. Stereoisomers
- II. Position isomers
- III. Chain isomers
- IV. Functional group isomers

Choose the **correct** answer from the options given below :

Options

1. A-I, B-IV, C-III, D-II
2. A-II, B-I, C-IV, D-III
3. A-III, B-I, C-IV, D-II
4. A-III, B-I, C-II, D-IV

Question Type : **MCQ**

Question ID : **860654889**

Option 1 ID : **8606543029**

Option 2 ID : **8606543030**

Option 3 ID : **8606543031**

Option 4 ID : **8606543028**

Q.63

Consider the following data :

$$\Delta_f H^\ominus (\text{methane, g}) = -X \text{ kJ mol}^{-1}$$

$$\text{Enthalpy of sublimation of graphite} = Y \text{ kJ mol}^{-1}$$

$$\text{Dissociation enthalpy of } H_2 = Z \text{ kJ mol}^{-1}$$

The bond enthalpy of C–H bond is given by :

Options

1. $X + Y + Z$
2. $\frac{-X + Y + Z}{4}$
3. $\frac{X + Y + 4Z}{2}$
4. $\frac{X + Y + 2Z}{4}$

Question Type : MCQ

Question ID : 860654879

Option 1 ID : 8606542988

Option 2 ID : 8606542991

Option 3 ID : 8606542990

Option 4 ID : 8606542989

Q.64

Given below are two statements :

Statement I : The correct order in terms of bond dissociation enthalpy is $Cl_2 > Br_2 > F_2 > I_2$.

Statement II : The correct trend in the covalent character of the metal halides is $[SnCl_4 > SnCl_2]$, $[PbCl_4 > PbCl_2]$ and $[UF_4 > UF_6]$.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. **Statement I** is true but **Statement II** is false
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is false but **Statement II** is true
4. Both **Statement I** and **Statement II** are false

Question Type : MCQ

Question ID : 860654883

Option 1 ID : 8606543006

Option 2 ID : 8606543004

Option 3 ID : 8606543007

Option 4 ID : 8606543005

Q.65 Given below are two statements :

Statement I : Crystal Field Stabilization Energy (CFSE) of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ is greater than that of $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$.

Statement II : Potassium ferricyanide has a greater spin-only magnetic moment than sodium ferrocyanide.

In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. Both **Statement I** and **Statement II** are true
2. **Statement I** is false but **Statement II** is true
3. **Statement I** is true but **Statement II** is false
4. Both **Statement I** and **Statement II** are false

Question Type : **MCQ**

Question ID : **860654885**

Option 1 ID : **8606543012**

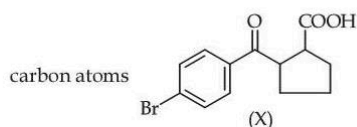
Option 2 ID : **8606543015**

Option 3 ID : **8606543014**

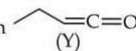
Option 4 ID : **8606543013**

Q.66 Given below are two statements :

Statement I : Compound (X), shown below, dissolves in NaHCO_3 solution and has two chiral



Statement II : Compound (Y), shown below, has two carbons with sp^3 hybridization, one carbon with sp^2 and one carbon with sp hybridization



In the light of the above statements, choose the **correct** answer from the options given below :

Options

1. Both **Statement I** and **Statement II** are false
2. Both **Statement I** and **Statement II** are true
3. **Statement I** is true but **Statement II** is false
4. **Statement I** is false but **Statement II** is true

Question Type : **MCQ**

Question ID : **860654888**

Option 1 ID : **8606543025**

Option 2 ID : **8606543024**

Option 3 ID : **8606543026**

Option 4 ID : **8606543027**

- Q.67** The correct statements are :
- A. Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis.
 - B. During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.
 - C. Nucleotides are joined together by glycosidic linkage between C_1 and C_4 carbons of the pentose sugar.
 - D. Quaternary structure of proteins represents overall folding of the polypeptide chain.
- Choose the correct answer from the options given below :

Options

- 1. A, B and D Only
- 2. B and C Only
- 3. A and B Only
- 4. A, C and D Only

Question Type : MCQ

Question ID : 860654894

Option 1 ID : 8606543049

Option 2 ID : 8606543050

Option 3 ID : 8606543048

Option 4 ID : 8606543051

- Q.68** Decomposition of A is a first order reaction at T(K) and is given by $A(g) \longrightarrow B(g) + C(g)$.

In a closed 1 L vessel, 1 bar A(g) is allowed to decompose at T(K). After 100 minutes, the total pressure was 1.5 bar. What is the rate constant (in min^{-1}) of the reaction ? ($\log 2 = 0.3$)

Options

- 1. 6.9×10^{-2}
- 2. 6.9×10^{-4}
- 3. 6.9×10^{-1}
- 4. 6.9×10^{-3}

Question Type : MCQ

Question ID : 860654881

Option 1 ID : 8606542996

Option 2 ID : 8606542999

Option 3 ID : 8606542997

Option 4 ID : 8606542998

Q.69 Aqueous HCl reacts with $\text{MnO}_2(\text{s})$ to form $\text{MnCl}_2(\text{aq})$, $\text{Cl}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$. What is the weight (in g) of Cl_2 liberated when 8.7 g of $\text{MnO}_2(\text{s})$ is reacted with excess aqueous HCl solution ? (Given Molar mass in g mol^{-1} Mn = 55, Cl = 35.5, O = 16, H = 1)

Options

1. 7.1
2. 14.2
3. 71
4. 21.3

Question Type : MCQ

Question ID : 860654876

Option 1 ID : 8606542978

Option 2 ID : 8606542976

Option 3 ID : 8606542979

Option 4 ID : 8606542977

Q.70 Given below are four compounds :

- | | |
|------------------------|-------------------------|
| (a) n-propyl chloride | (b) iso-propyl chloride |
| (c) sec-butyl chloride | (d) neo-pentyl chloride |

Percentage of carbon in the one which exhibits optical isomerism is :

Options

1. 46
2. 52
3. 56
4. 40

Question Type : MCQ

Question ID : 860654890

Option 1 ID : 8606543034

Option 2 ID : 8606543032

Option 3 ID : 8606543033

Option 4 ID : 8606543035

Section : Chemistry Section B

Q.71 Identify the metal ions among Co^{2+} , Ni^{2+} , Fe^{2+} , V^{3+} and Ti^{2+} having a spin-only magnetic moment value more than 3.0 BM. The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is _____.

Question Type : SA

Question ID : 860654900

Q.72 The osmotic pressure of a living cell is 12 atm at 300 K. The strength of sodium chloride solution that is isotonic with the living cell at this temperature is _____ g L^{-1} . (Nearest integer)

Given : $R = 0.08 \text{ L atm K}^{-1} \text{ mol}^{-1}$

Assume complete dissociation of NaCl

(Given : Molar mass of Na and Cl are 23 and 35.5 g mol^{-1} respectively.)

Question Type : SA

Question ID : 860654897

Q.73 A substance 'X' (1.5 g) dissolved in 150 g of a solvent 'Y' (molar mass = 300 g mol^{-1}) led to an elevation of the boiling point by 0.5 K. The relative lowering in the vapour pressure of the solvent 'Y' is $\text{_____} \times 10^{-2}$. (nearest integer)

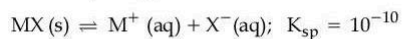
[Given : K_b of the solvent = $5.0 \text{ K kg mol}^{-1}$]

Assume the solution to be dilute and no association or dissociation of X takes place in solution.

Question Type : SA

Question ID : 860654896

Q.74 MX is a sparingly soluble salt that follows the given solubility equilibrium at 298 K.



If the standard reduction potential for $\text{M}^+ \text{(aq)} \xrightarrow{+e^-} \text{M(s)}$ is $\left(E_{\text{M}^+/\text{M}}^\ominus\right) = 0.79 \text{ V}$, then the value of

the standard reduction potential for the metal/metal insoluble salt electrode $\text{E}_{\text{X}^-/\text{MX (s)}/\text{M}}^\ominus$ is _____ mV . (nearest integer)

[Given : $\frac{2.303 RT}{F} = 0.059 \text{ V}$]

Question Type : SA

Question ID : 860654899

Q.75 The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in 0.1 M H_2X solution is $\text{_____} \times 10^{-15} \text{ M}$. (Nearest Integer)

Question Type : SA

Question ID : 860654898